



CARBON AND ITS VARIOUS FORMS — FULL MIND MAP

Chemistry · Chapter 6.A · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / State PCS

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1. Carbon Basics, Catenation & Major Allotropes

- Carbon sabse zyada compounds banane wala element mana jata hai, mainly uski catenation (apne atoms ke saath long chains/rings banana) aur tetravalency ki wajah se. Diamond, Graphite aur Coal teeno carbon-containing forms/materials hain. [Q1, Q2, Q3]
- Thermodynamically Carbon ka sabse stable allotrope Graphite hai. Diamond aur Graphite dono pure Carbon se bane hote hain, lekin unki crystal structure, hardness aur electrical conductivity bahut alag hoti hai. [Q4, Q5, Q6]
- Diamond 3D tetrahedral covalent network banata hai aur electrical insulator hota hai; Graphite layered structure wala soft material hai jisme delocalised electrons ki wajah se electrical conduction hoti hai. [Q5, Q6]
- Diamond, Graphite aur Fullerene pure Carbon ke allotropes hain; Oxocarbon koi allotrope nahi, balki Carbon + Oxygen se bana compound class hai. [Q12]

2. Lab-grown Diamond, Graphene & Modern Carbon Materials

- Lab-grown diamonds mainly HPHT aur CVD methods se banaye jate hain. Source questions Graphite ko Carbon source/seed-base ke roop mein key karte hain; high temperature-pressure conditions mein Carbon diamond structure mein crystallise kar sakta hai. [Q7, Q8]
- Chronology recall: Kevlar (1965) → Fullerene (1985) → Graphene isolation (2004) → TIFR ka nano-gold based 'black gold' work (2019). Is source question ka sequence 4-2-3-1 hai. [Q9]
- Graphene pure Carbon ka one-atom-thick, two-dimensional nano-structure/allotrope hai; Kevlar, Lexan aur spider silk pure Carbon-only materials nahi hain. [Q10, Q11]

⚠ EXAM TRAP — Lab-grown diamond mein 'seed' aur 'carbon source' ko mix mat karo

- ▶ Practical CVD/HPHT growth mein actual seed usually ek chhota diamond crystal/wafer hota hai; Graphite ya hydrocarbon feed Carbon source ho sakta hai.
- ▶ Source Q7–Q8 Graphite ko seed/base wording mein key karta hai—exam mein source wording dekhna, conceptually seed crystal aur carbon feed alag cheezein hain.

3. Pencil Lead & Carbon Fibre

- Pencil ka 'lead' actual Lead metal nahi hota; woh Graphite (often clay ke saath mixture) hota hai. Isi wajah se pencil writing graphite deposition se hoti hai. [Q13, Q14]
- Carbon-fibre reinforced composites lightweight aur high-strength hote hain, isliye automobiles, racing cars, aircraft aur spacecraft parts mein use hote hain. Carbon-fibre waste ko mechanical, thermal ya chemical routes se recycle kiya ja sakta hai. [Q15]

4. Fullerene / C_{60} — Discovery, Structure & Matching

- Fullerene discovery ke liye 1996 Chemistry Nobel Robert Curl, Harold Kroto aur Richard Smalley ko mila; Richard Feynman is discovery team ka hissa nahi the. [Q16]
- Buckminsterfullerene C_{60} ek hollow cage/soccer-ball type molecule hai jisme 60 Carbon atoms pentagons aur hexagons wale polyhedral network mein arranged hote hain. [Q18]
- Fullerene mein Fluorine nahi hota. Isliye 'Fullerene = fluorine-containing organic compound' wali matching galat hai; ye pure Carbon allotrope hai. [Q17, Q19, Q20, Q21]
- Matching recall: Dry ice = solid CO_2 ; Mustard gas = toxic chemical warfare agent; Freon = refrigerant; SO_2 acid rain ka major contributor hai. [Q17, Q19, Q20, Q21]

⚠ EXAM TRAP — Fullerene ≠ Fluorine compound

- ▶ Naam similar lagne ki wajah se confusion hota hai, lekin Fullerene pure Carbon allotrope hai.
- ▶ Fluorine-containing polymer ka classic example Teflon (PTFE) hai, Fullerene nahi.

5. Carbon in Coal, Lignite & Iron Products

- Sand (mainly silica, SiO_2) mein Carbon nahi hota, jabki Diamond, Graphite aur Coal carbon-containing hain. Lignite bhi low-grade coal hai aur Carbon contain karta hai. [Q22, Q23]
- Coal rank ke saath Carbon percentage generally badhta hai: Peat < Lignite < Bituminous < Anthracite; in options Anthracite mein Carbon sabse zyada hota hai. [Q24]
- Iron products mein Pig Iron ka Carbon content comparatively sabse high (~3.5–4.5%) hota hai; Wrought Iron bahut low-carbon hota hai aur Steel intermediate range mein aata hai. [Q25]

6. Crystal, Glass Etching & Carat

- Diamond, Quartz aur Graphite crystalline solids hain. Source ke historical question mein Sulphur ko 'not in crystal form' key kiya gaya hai, although Sulphur ke crystalline allotropes bhi known hain. [Q26]
- Glass ko Diamond scratch kar sakta hai aur Hydrofluoric acid (HF) chemically etch karta hai kyunki HF silica (SiO_2) ke saath react karta hai. Aqua regia ya conc. H_2SO_4 is standard glass-etching answer nahi hain. [Q27]
- Gemstones/diamonds ke mass ke liye metric carat use hota hai: 1 carat = 200 mg = 0.2 g. [Q28]

⚠ EXAM TRAP — Q26 source-specific hai

- ▶ Elemental Sulphur ke rhombic aur monoclinic crystalline allotropes hote hain; isliye 'Sulphur crystalline nahi hota' ko universal chemistry fact mat samajhna.
- ▶ Is question ko historical/source-key based odd-one-out ke roop mein treat karo.

7. Graphene — Strength, Conductivity & Uses

- Graphene Carbon ka 2D, one-atom-thick sheet hai; excellent electrical conductivity, extreme thinness aur very high tensile strength uski key properties hain. [Q29, Q30]
- Graphene Silicon ka allotrope nahi hai. Transparent-conducting nature ki wajah se touchscreens, LCD/OLED electrodes aur advanced electronics mein iska potential bahut high hai. [Q29]

⚠ EXAM TRAP — 'Strongest material' ka exam shorthand

- ▶ Source options mein Graphene correct hai, lekin 'strongest' property ko tensile strength, hardness aur toughness se confuse mat karo—ye alag material properties hain.

8. Graphite as Lubricant + Mixed Chemistry Recall

- Permanent hard water mein dissolved $\text{CaSO}_4/\text{MgSO}_4$ jaise salts ho sakte hain. Diamond Copper aur Iron se much harder hai; atmosphere ka main component Nitrogen (~78%) hai aur vegetable ghee hydrogenation mein Hydrogen use hota hai. [Q31]
- Graphite solid lubricant hai. Layered structure mein sheets ek-dusre par easily slide karti hain, isliye heavy machinery aur high-temperature/high-pressure lubrication applications mein useful hota hai. [Q32, Q33]

9. Dry Ice — Solid CO_2

- Dry Ice = solid/frozen Carbon dioxide (CO_2). Ye ordinary water ice nahi hota aur cooling agent ke roop mein use hota hai. [Q34, Q35, Q36, Q37, Q38]
- Dry Ice atmospheric pressure par melt hone ke bajay directly solid se gas mein sublime karta hai; isi wajah se ise 'dry' ice kaha jata hai kyunki liquid-water residue nahi chhodta. [Q34–Q38]

10. Water Gas, Soda Water & CO Bond Order

- Water gas ek synthesis-gas mixture hai: $\text{CO} + \text{H}_2$. Iska use methanol aur dusre industrial chemicals ke synthesis mein kiya ja sakta hai. [Q39]
- Soda-water bottle kholne par pressure kam hota hai aur dissolved CO_2 bubbles ke roop mein bahar nikalta hai; released gas Carbon dioxide hoti hai. [Q40]
- Molecular Orbital Theory ke hisab se bond order = $\frac{1}{2}(\text{Nb} - \text{Na})$. CO ke liye effective bond order 3 hota hai, jo strong triple-bond character ko indicate karta hai. [Q41]

✓ Source-question mapping preserved: Q1–Q41